T_7

$$T_{6+1} = \binom{9}{6} x^{3} \cdot y^{6}$$

$$T_{6+1} = \binom{9}{6} \left(\frac{x}{3}\right)^{3} \cdot \left(\frac{-3}{2\sqrt{x}}\right)^{6}$$

$$T_{6+1} = \frac{\overset{3}{\cancel{9}} \cdot \overset{4}{\cancel{8}} \cdot 7}{\cancel{3} \cdot \cancel{2}} \cdot \frac{\cancel{x^{3}}}{3^{3}} \cdot \frac{-3^{6}}{2^{6}\, \cancel{x^{3}}}$$

$$T_{6+1} = 7 \cdot \frac{-3^{4}}{2^{4}}$$

$$T_7 = \frac{7 - 3^{4}}{2^{4}}$$

b) solo emfocando solo en x

$$T_{k+1} = \binom{9}{k} \left(\frac{x}{3}\right)^{9-k} \left(-\frac{3}{2\sqrt{x}}\right)^{k}$$

$$T_{2+1} = x^{9-k} \cdot x^{-\frac{k}{2}}$$

$$9 - \frac{3k}{2} = 7$$

$$-3k = -4$$

$$k = \frac{4}{3}$$

$$T_5 = \binom{9}{4}\left(\frac{x}{3}\right)^{9-4}\left(-\frac{3}{2\sqrt{x}}\right)^4 = \frac{126}{3^6}\cdot\frac{3^4}{2^4}\cdot\frac{x^5}{x^2} = \frac{126}{3\cdot 16}x^3$$

$$T_6 = \binom{9}{5}\left(\frac{x}{3}\right)^{9-5}\left(-\frac{3}{2\sqrt{x}}\right)^5 = -\frac{126}{3^4}\cdot\frac{3^5}{2^5}\cdot\frac{x^4}{x^{5/2}} = -\frac{126\cdot 3}{32}x^{4-\frac{5}{2}}$$

* Sumar coeficientes

2) $T_{K+1} = \binom{7}{K}(1)^{7-K}\left(\frac{2}{x}\right)^K$

$x^{-K} = -1$

$-K = -1$

$K = 1$

$T_2 = x^{-1}$

Cof de T_2 es : $7\cdot 2 = 14$

$x^{1-K} = x^0$

$1-K=0$

$K=1$

$T_1 = \binom{7}{0}\cdot\left(\frac{2}{x}\right)^0$

$T_1 = 1$

3) $S = \{ n \in \mathbb{N} : \sum_{i=1}^{n} \frac{1}{(2i-1)(2i+1)} = \frac{n}{2n+1} \}$

$1 \in S? = \sum_{i=1}^{1} \frac{1}{3} = \frac{1}{3}$ ✓

$H.I: K \in S \Rightarrow \sum_{i=1}^{K} \frac{1}{(2i-1)(2i+1)} = \frac{K}{2K+1}$

$T.I: K+1 \in S \Rightarrow \sum_{i=1}^{K+1} \frac{1}{(2i-1)(2i+1)} = \frac{K+1}{2K+3}$

$$\sum_{i=1}^{K+1} \frac{1}{(2i-1)(2i+1)} = \sum_{i=1}^{K} \frac{1}{(2i-1)(2i+1)} + \frac{1}{(2(K+1))(2(K+1)+1)} \Rightarrow \frac{K}{2K+1} + \frac{1}{(2K+1)(2K+3)}$$

Probar: $\frac{K(2K+3)+1}{(2K+1)(2K+3)} = \frac{2K^2+3K+1}{(2K+1)(2K+3)} \Rightarrow \frac{(2K+1)(K+1)}{(2K+1)(2K+3)} = \frac{K+1}{2K+3}$

$\therefore$ Por lo tanto P_{K+1}, es V, $K+1 \in S$, Así $S = \mathbb{N}$

$S = \{ m \in \mathbb{N} : 3^{2m} + 7$ es div Por 8$\}$

$1 \in S$? $3^2 + 7 \Rightarrow 16$ ✓

H.I, $K \in S \Rightarrow 3^{2K} + 7$; 8P

T.I, $K+1 \in S \Rightarrow 3^{2K+2} + 7$

ComProbamos: $3^{2K} \cdot 3^2 + 7$

$$3^{2K} \cdot 9 + 7$$

$$9^K \cdot 9 + 7$$

$$9^K \cdot 9 + 63 - 56$$

$$9\underbrace{(9^K + 7)}_{H\,I} \underbrace{- 56}_{\text{Div Por } 8}$$

$\therefore$ Por lo tanto P_{K+1}, es verdad ; $K+1 \in S$